AYUSH KULSHRESTHA

UPSTREAM ENERGY & AI PROFESSIONAL | SUBSEA INTELLIGENCE | ML ENGINEERING

ayushkul404@yahoo.com | +91 8826206605 |

www.akenergyconsultant.org

LinkedIn | GitHub

6+	98%	90%	1000+	$2-5M
YEARS EXPERIENCE	ML ACCURACY	PROCESS REDUCTION	ASSETS MANAGED	CLIENT SAVINGS

🎯 OBJECTIVE

Upstream Energy & AI Professional with 6+ years at Wood Mackenzie and KPMG, combining deep subsea production expertise (Christmas Trees, SURF systems, FPSOs, manifolds, topsides, wellheads) with hands-on AI/ML engineering. Built production-grade ML systems (PyTorch, 3.5M parameters, 98% accuracy) that automate upstream contract intelligence, and Python automation pipelines that reduced manual processes by 90%. Expert in subsea contract lifecycle tracking (Concept → Pre-FEED → FEED → EPC → Installation → Hook-up & Commissioning), upstream production databases, and AI-driven market forecasting. Seeking hybrid roles at the intersection of upstream oil & gas strategy and applied AI/ML.

Visit my portfolio: www.akenergyconsultant.org to explore detailed case studies and project showcases.

🎓 EDUCATION

MBA (Marketing and International Business)	Galgotias College of Engineering & Technology	**2018 - 2020**
B. Tech (Mechanical Engineering)	Maharshi Dayanand University	**2013 - 2017**

💼 WORK EXPERIENCE

Wood Mackenzie, Gurugram

2024 - Present

Upstream Production Analyst & AI Developer (Oil & Gas)

SUBSEA INTELLIGENCE & CONTRACT TRACKING

- Tracks 500+ subsea contracts across full lifecycle (Concept → Pre-FEED → FEED → EPC → Installation → Hook-up & Commissioning → O&M → Decommissioning) covering Christmas Trees, SURF packages, manifolds, FPSOs, jackets, topsides, and pipelines across global deepwater basins.
- Executes quarterly analyses of SURF contract awards, Christmas Tree procurement, and manifold installations — translating complex subsea intelligence into actionable advisory deliverables for C-suite decision-makers.
- Managed the global rig and USC (Upstream Supply Chain) database covering 1,000+ active drilling assets and subsea infrastructure (FPSOs, jackets, pipelines, wellheads) — providing clients with real-time intelligence on rig availability, dayrates, contract status, and EPC award pipelines.
- Advised E&P operators on optimal contract timing by analyzing historical EPC award patterns, vessel availability, and yard capacity — saving clients an estimated $2-5M per project in avoided schedule overruns.

AI/ML & AUTOMATION

- Deployed AI-driven anomaly detection across subsea contract datasets — flagging data inconsistencies and market shifts 2 weeks ahead of manual review cycles, enabling proactive client alerts on SURF award pipeline changes.
- Designed and implemented Python automation pipelines replacing 40+ hours/week of manual data processing — reducing report delivery time from 2 weeks to 2 days (90% reduction) and freeing analysts for higher-value client advisory work.
- Built autonomous ML-powered contract classification system processing 2,800+ articles weekly with 98% accuracy — ensuring zero missed high-priority EPC/FEED awards worth $100M+.
- Re-architected data delivery workflows with self-service dashboards for on-demand market views, reducing manual touchpoints by 60% and cutting ad-hoc request volume by half.

CLIENT ADVISORY

- Coordinates primary research across 6 global offices and clients to close critical data gaps — delivering consulting-grade upstream intelligence that directly feeds into client strategy.
- Served as first point of escalation for strategic accounts, resolving data discrepancies within 24-hour SLAs and maintaining 98%+ client satisfaction across quarterly engagement reviews.

- Partnered with senior consultants to structure client deliverables on subsea market outlook, contract benchmarking, and equipment demand forecasting.

🚀 Engineered a full-stack analytics platform revolutionizing offshore drilling decision-making:

Python (7,000+ lines) | Streamlit | ML | Monte Carlo | NLP

Challenge: Oil & gas operators lacked integrated intelligence systems for rig efficiency analysis and climate risk assessment, leading to $2-5M cost overruns per project.

Solution: Built enterprise-grade Streamlit app with 15+ AI modules:

- Climate intelligence engine covering 30+ regions
- ML-based rig-well matching algorithm (95% accuracy)
- Monte Carlo simulators for project risk quantification
- NLP chatbot for natural language data queries
- Real-time efficiency scoring with anomaly detection
- 50+ interactive dashboards for contract negotiations

✓ Impact: Slashed contractor evaluation time from 8 hours to 15 minutes

▶ Live Demo

🤖 Built an AI-powered contract intelligence agent for upstream news monitoring:

PyTorch (3.5M params) | Flask + SocketIO | TextRank | Regex+ML

Challenge: Upstream analysts manually review 2,800+ articles weekly for contract opportunities — spending 15+ hours on classification with risk of missed high-priority EPC/FEED awards worth $100M+.

Solution: Built production-ready Python application (10,000+ lines):

- PyTorch neural network classifier (3.5M parameters, 98% accuracy)
- Hybrid regex+ML contract detection engine
- Autonomous background agent with desktop alerts
- TextRank extractive summarization
- Flask + SocketIO real-time dashboard with keyboard-driven workflow

✓ Impact: 70% time reduction | 169 companies tracked | Zero missed contracts

▶ GitHub Repo

KPMG Global Services, Gurugram

2022 - 2024

Business Associate (Energy & Natural Resources, Auto and IM)

- Crafted visually engaging Power BI dashboards, transforming complex data into actionable insights to support strategic decision-making within the Energy and Natural Resources (ENR) sector.
- Conducted thorough data analysis by utilizing VBA and Python for querying and data cleaning, ensuring data accuracy and facilitating informed decision-making processes.
- Applied expertise in pricing and strategy, including market share analysis, price evaluation, and peer comparison, to inform pricing strategies for ENR clients and optimize their market positioning.
- Led benchmarking initiatives to provide comprehensive insights into industry rates and pricing dynamics, aligning service rates with industry standards and enhancing competitiveness.
- Leveraged advanced statistical and text mining techniques to extract actionable insights from large datasets, driving informed decision-making and business impact.
- Utilized Python and VBA for data manipulation and analysis, automating processes to improve efficiency and enhance modeling accuracy.
- Applied agile methodologies to identify and implement technology solutions, driving efficiency and innovation in data management processes within the ENR sector.

Generative AI Research Automation:

Excel VBA | Generative AI API Integration

Situation: Team was struggling with manually gathering data for secondary research projects, leading to inefficiencies and errors.

Task: Find a solution to streamline data gathering, reduce time on manual research, and improve accuracy.

Action: Developed an Excel VBA macro integrated with a cutting-edge generative AI tool, automating a significant portion of the secondary research process.

✓ Result: Marked reduction in data gathering time, enhanced accuracy, and more efficient research process.

ACS Global Tech Solutions, Noida

2020 - 2022

Market Research Associate (Information Technology in Industrial Manufacturing)

- Market Analysis and Strategy Development: Proficient in understanding market dynamics, competitor landscape, and industry trends to inform strategic decision-making.
- Competitive and Trend Analysis: Conducts competitive and trend analysis to provide actionable insights, translating key trends into strategic recommendations.
- Product Mapping and Competitor Performance Tracking: Focuses on understanding market positioning and identifying areas for strategic improvement.

- Content Creation and Thought Leadership: Showcases manufacturing expertise through authored blogs on digital manufacturing technologies.
- Utilization of Databases and Market-Related Product Data: Proficient in using databases for information gathering and market analysis.

Continental Automotive Components, Gurugram — 2019

Trainee

- Gained insights into the operation and functions of cluster units.
- Acquired knowledge of cluster unit manufacturing processes.
- Learned about quality control procedures for cluster units.

TECHNICAL SKILLS & CERTIFICATIONS

AI / Machine Learning	PyTorch (3.5M parameters), Neural Networks, NLP (TextRank, Classification), Monte Carlo Simulation, Anomaly Detection, Generative AI, Feature Engineering, Model Training & Deployment, scikit-learn — 98% classification accuracy
Programming	Python (10,000+ lines production code), VBA, Flask, Streamlit, SocketIO, SQL, Pandas, NumPy, Data Pipelines, ETL
Upstream Domain	SURF Systems, Christmas Trees, Manifolds, FPSOs, FLNGs, FSRUs, Wellheads, Topsides, Jackets, Subsea Controls, Contract Lifecycle (Concept → Decommissioning), EPC Tracking
Data & BI	Power BI, Excel Dashboards, Google Analytics, Microsoft Office (Excel, PowerPoint, Word), Statistical Analysis, Data Modeling, Tableau
Energy Databases	Refinitiv, FactSet, Bloomberg, GlobalData, Zoominfo
Product & Research	Product Roadmap, Stakeholder Management, KPIs, Cross-functional Collaboration, Market Sizing (TAM/SAM/SOM), Competitive Intelligence, Primary & Secondary Research, SWOT Analysis
Methodologies	Lean Six Sigma, Agile, Data Pipeline Architecture, Jira
Certifications	Lean Six Sigma (KPMG) • Fundamentals of Digital Marketing (Google) • Business Analysis: Essential Tools & Techniques (IIBA) • Python Essential Training (LinkedIn Learning) • Digital Economy for Sustainable Growth (ADBI) • Growth & Inclusion in Asia's Cities (ADBI)
Languages	English, Hindi